10th International Symposium on Test Automation & Instrumentation (ISTAI 2024) Shenzhen, China 14–16 August 2024

# KNOWLEDGE GRAPH CONSTRUCTION FOR CROSS ENTERPRISE PRODUCT SUPPLY RELATIONSHIP OF INTELLIGENT INSTRUMENTS

***Min Xu[1], Min Huang[1*], Yiming Li[1]***

*[1]School of Mechanical Electrical Engineering, Beijing Information Science and Technology University, Beijing, China*
**huangmin@bistu.edu.cn*

**Keywords**: INTELLIGENT INSTRUMENTS, COLLABORATIVE MANUFACTURING, SUPPLY RELATIONSHIP, KNWOLEDGE GRAPH

## Abstract

The massive resources of intelligent instruments provide data support for cross enterprise regional network collaborative manufacturing. However, the discrete distribution of data, a large amount of redundancy, and weak correlation limit system knowledge learning. This paper proposes an automatic construction method of a knowledge graph for the supply and demand matching of products and the relationship between upstream and downstream enterprises. Firstly, Stanford CoreNLP tool is used to achieve enterprise entity recognition. Secondly, considering the potential relationship between boundary words and product keywords, an improved CRF model is proposed to achieve product entity recognition. Then, a method for extracting enterprise products supply relationships based on relationship indicator lexicon and dependency syntax analysis is proposed to obtain the upstream and downstream relationships of enterprise products. Finally, the Neo4j database is used to store the knowledge graph of cross enterprise product supply relationship of intelligent instruments, and it is published on a third-party service platform to promote efficient resource sharing and collaborative manufacturing of intelligent instruments across enterprises.

## 1 Introduction

The information exchange and coordination of numerous businesses between upstream and downstream enterprises of instruments are important contents in achieving informatization in the instrument manufacturing industry. However, due to the large number and variety of manufacturing resources, the wide geographical distribution, the large proportion of unstructured data, and the complexity of production information systems within many enterprises. Adequately extracting rich semantic information between various manufacturing resources and fully expressing the association between multimodal and multi-source data becomes the key to intelligent web-based collaborative manufacturing. However, there are numerous and diverse manufacturing resources, with a wide geographical distribution, a large proportion of unstructured data, and complex production information systems within many enterprises. Fully extracting rich semantic information between various manufacturing resources and fully expressing the correlation between multimodal and multi-source data has become the key to intelligent network collaborative manufacturing. Knowledge graph is a new generation of intelligent search engine technology that can achieve knowledge storage, relationship acquisition, and knowledge expression functions. It performs well in multiple downstream tasks and is currently an important research direction. Knowledge storage and semantic relationship acquisition are the primary functions of knowledge graphs, which can be used to express resource relationships and knowledge content within a certain field[1-3]. Wen et al.[4] proposed an intelligent manufacturing resource process knowledge graph to address the limitations of manufacturing process planning that heavily relies on manual experience, and applied it to process planning scenarios. Jiang et al. analyzed the dynamic time series of data in the remanufacturing disassembly process and established a dynamic data flow remanufacturing disassembly process knowledge graph, providing data support for the remanufacturing disassembly process planning method. The knowledge graph also demonstrates good performance in the application layer of intelligent retrieval and inference. Due to its enhanced semantic expression ability of the engine, the system is easier to obtain potential contextual relationships and can better understand the user's intentions[5-7]. Cai et al.[8] proposed a knowledge graph driven diagnostic method for the application scenario of rotating machinery fault diagnosis. Based on the state relationship of multi-source data and the structural relationship of mechanical equipment, a knowledge base was established, and machine learning training was used to improve the accuracy of fault diagnosis. Wu et al.[9] proposed an intelligent diagnosis model for industrial equipment faults based on multimodal knowledge graphs, which combines intelligent target recognition algorithms to map feature vectors to entity, attribute, and relationship vectors in the knowledge graph, and perform semantic matching inference based on this mapping.

Knowledge graphs have been extensively studied in the manufacturing industry. However, there is relatively little research on the application in the field of intelligent instruments, especially in the cross enterprise product supply

https://doi.org/10.1049/icp.2024.2707 

and demand matching of the intelligent instrument industry. Therefore, this paper proposes an automatic construction method for cross enterprise product relationship knowledge graph of intelligent instruments, with the main contributions as follows:

(1) A method of enterprise entity identification and product entity identification is proposed based on the manufacturing resource information of intelligent instruments. Stanford CoreNLP tool is used to realize enterprise entity identification, and product entity identification is carried out by using a combination of boundary words, product keywords and CRF models.

(2) Combining corpus features and rule post-processing methods, a method for extracting enterprise supply relationships based on relationship indicator lexicon and dependency syntax analysis is proposed.

(3) A knowledge graph of cross enterprise product supply relationship for intelligent instruments is constructed and published on a third-party service platform.

The remaining content of this paper is organized as follows: Section 2 describes the method of entity identification for enterprises and products. Section 3 describes the extraction of enterprise product relationships. Section 4 describes the construction and publication of the knowledge graph. Section 5 is conclusion.

## 2. Entity Identification of Enterprises and Products

*2.1 Enterprise Entity Identification*

Enterprise entity recognition has achieved the extraction of enterprise names from unstructured text. This paper uses the Stanford CoreNLP entity identification toolkit, which integrates named entity parsing, part of speech tagging, sentiment analysis, syntactic analysis, word segmentation, and other functions. Moreover, Stanford CoreNLP can meet the accuracy and recall requirements for enterprise entity recognition. Therefore, this study uses Stanford CoreNLP to recognize enterprise names[10]. Firstly, preprocessing is performed to remove characters such as spaces and punctuation from the input text vector. Then, the text is segmented according to its structure to obtain a phrase sequence. Finally, special enterprise names are identified.

*2.2 Product Entity Identification*

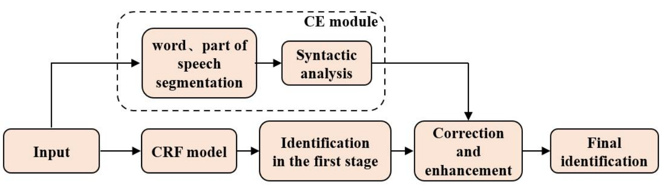


Fig. 1 Product entity identification processing flow

Taking into full consideration the boundary words and enterprise product keyword features, based on the first stage identification of the Conditional Random Field (CRF) model, product entity groups are obtained through part of speech segmentation and dependency syntax analysis to enhance and modify the CRF model results and improve the identification performance of complex products. The process is shown in Figure 1.

*2.2.1 Implementation of CRF Model:* The CRF model is trained in a supervised manner. In order to solve the problem of limited sample size in intelligent instrument manufacturing resources, data augmentation is performed on entities of similar types in the domain dictionary. The Jieba segmentation tool[11] is used to segment existing text and perform analogical substitution on entities, generating a new piece of data to expand the dataset, as shown in Figure 2.


Fig. 2 Expanding data by replacing keywords

The raw data of the training corpus comes from the official websites of various instrument companies, and the identification model is established using the CRF toolkit. The BIEO method is used to label entity categories. The left boundary label of the product name is B_PRODUCT, the right boundary label of the product name is E_RODUCT, the middle character label of the left and right boundaries is I_PRODUCT, and the label of other non-product name words is O, as shown in Table 1.

Table 1 Entity category annotation

| Label | Value |
|---|---|
| B_PRODUCT | Left boundary |
| E_PRODUCT | Right boundary |
| I_PRODUCT | Middle character |
| O | Non product name |

The corpus is the foundation for implementing and testing various natural language processing methods, and 500 statements related to enterprise products are selected from the official website of instrumentation companies. With the addition of amplified data samples, the final training corpus includes 1005 product category words. The training and testing dataset is stored in a multi row format, with each row is a token containing rich semantic features. Feature channels are separated by spaces or tabs, and category labels are vector tail features. After annotation, the output is converted to a format that meets the requirements of CRF++, and used as input for training.

*2.2.2 CE Module:* The features of corpus mainly include word features, type features, boundary word features, and dictionary features. Among them, word features represent the current word or a neighbor phrase, and adding numbers represents the relative distance between the current word and other words; The type features include letters, numbers, punctuation marks, special characters, etc; Boundary word features describe the situation where certain nouns frequently appear near a certain noun; Dictionary features are used to determine whether tokens belong to the preset dictionary. The comprehensive utilization of these features helps to improve the performance and accuracy of named entity identification systems.

Syntactic analysis mainly involves rule post-processing, which takes sentences as the basic unit and obtains product entity groups through syntactic analysis. Taking the processing flow of the statement “Enterprises preparing to purchase electronic product series such as photoelectric sensors, laser detectors and power modules, lithium batteries, and switching devices” as an example, the two identified product entity groups are {photoelectric sensors, laser detectors} and {power modules, lithium batteries}. Partial syntactic analysis results are shown in Figure 3.

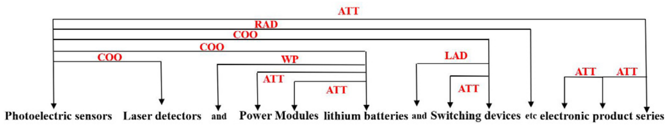


Fig. 3 Syntactic analysis results.

According to syntactic analysis, it can be concluded that photoelectric sensors, laser detectors, lithium batteries, and switching devices have a parallel relationship. Meanwhile, photoelectric sensors, laser detectors, and lithium batteries have been labeled as physical products. Based on this information, as switching devices appear side by side with other product entities, they should also be defined as one product entity with the same syntax structure and positional relationships. After labeling “lithium batteries” as products, define the left boundary “power supply” and “module” as having an ATT relationship (Attribute), and then splice them together to obtain the complete product name “power module”. The final product entity identification result is obtained by using the product entity group to correct the results of the first stage of the CRF model.

## 3 Extraction of Enterprise and Product Relationships

The extraction of the relationship between enterprises and products mainly focuses on the supply relationship between upstream and downstream enterprises, and its processing flow chart is shown in Figure 4.

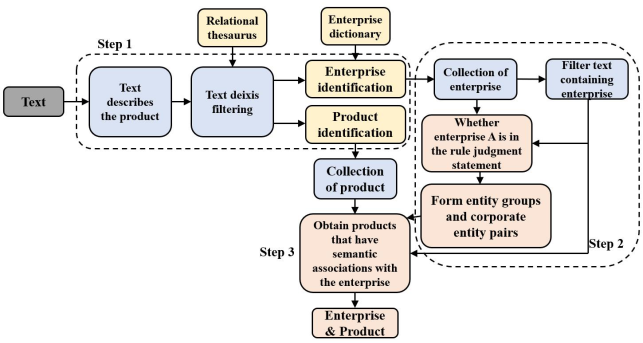


Fig. 4 Enterprise and product relationship extraction method flow.

Based on the results of enterprise entity identification and product entity identification, firstly, it is determined that the overall text is related to the enterprise supply relationship by traversing the relationship lexicon and enterprise dictionary, and the enterprise set and product set are divided from the enterprise and product identification results. Then generate enterprise entity pairs, determine suppliers, and obtain customer company and product corpus. Finally, determine the product information, obtain multiple enterprises or products that appear in the sentence through syntactic analysis, correspond and match customers with products, and finally determine the relationship chain between suppliers, customers, and products.

## 4 Construction of Cross Enterprise Product Supply Relationship Knowledge Graph

Integrate the obtained enterprise entities, product entities, and supply relationships between enterprise products to form the <Enterprise-Relationship-Product> triplet as the basic data format. The Neo4j graphical database is used to visualize and store knowledge nodes and edges, and complete the construction of an intelligent instrument enterprise upstream and downstream relationship knowledge graph, as shown in Figure 5. By establishing a knowledge graph of the upstream and downstream relationships of intelligent instrument enterprise products, the complex correlation relationships of enterprise products in the instrument industry can be effectively revealed. The construction of this correlation not only helps to gain a deeper understanding of the interactions and dependencies among various entities within the enterprise, but also provides deeper insights into the flow, production, and market positioning of products in the supply chain. At the same time, it can also help enterprises better manage their product lifecycle and supply chain relationships, and help identify potential cooperation opportunities and optimize overall business operations.

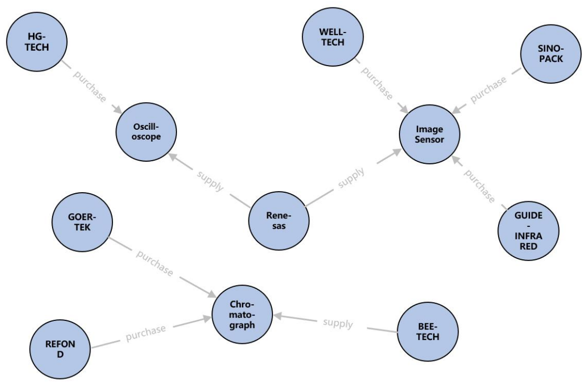


Fig. 5 Intelligent instrument cross enterprise product supply relationship knowledge graph.

Finally, an intelligent instrument regional network collaborative manufacturing service platform system was developed using HTML+JavaScript+CSS software architecture. The platform integrates industry chain service functions such as resource sharing, enterprise collaboration, expert consultation, and talent training. In the system, the intelligent instrument cross enterprise product supply relationship knowledge graph is published, which can promote information sharing between enterprises and help accelerate the efficiency of networked collaborative manufacturing between instrument enterprises, as shown in Figure 6.

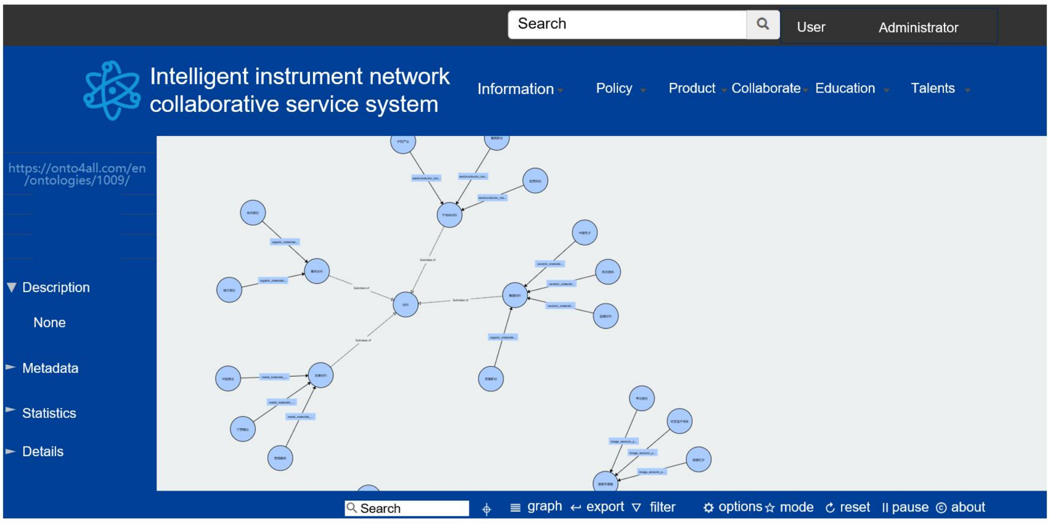


Fig. 6 Knowledge graph published on third-party service platform.

Publishing the knowledge graph on the third-party service platform has produced good results. For example, in the product planning stage, XX intelligent instrument enterprise obtained market demand and sales forecast data through the platform, and connected with the supply chain management system in real time for material demand planning and supplier selection. The enterprise directly accessed relevant data from the supply chain management system, such as suppliers' delivery times, stock levels and pricing information, in order to carry out rational procurement planning and supplier evaluation. The enterprise monitors production progress and quality data in real time through the platform. By connecting with third-party platforms, the enterprise can achieve remote monitoring and troubleshooting of equipment to ensure the stability and efficiency of the production process. In this application, the potential connection between products and enterprises is established through the knowledge graph, which helps to optimise the product supply chain of suppliers and customer enterprises and improve the flexibility and responsiveness of the supply chain. During the production process, the enterprise monitored the production progress and quality data in real-time through the platform. By connecting with third-party platforms, the enterprise achieved remote monitoring and fault diagnosis of equipment to ensure the stability and efficiency of the production process. In this application, the knowledge graph is used to establish potential connections between products and enterprises, helping to optimize the product supply chain of suppliers and customer enterprises, and improving the flexibility and responsiveness of the supply chain.

## 5 Conclusion

This paper proposes a method for enterprise and product entity recognition based on the supply and demand matching of intelligent instrument products and the relationship between upstream and downstream enterprises. The Stanford CoreNLP tool is used to achieve enterprise entity recognition. In addition, boundary words, product keywords, and CRF models are combined for product entity recognition. On this basis, considering the characteristics of the corpus, a method for extracting enterprise supply relationships based on relationship indicator lexicon and dependency syntax analysis is proposed through rule post-processing. Finally, a knowledge graph of cross enterprise product supply relationships for intelligent instruments is constructed and published on a third-party service platform to promote cross enterprise resource sharing and collaborative manufacturing in the intelligent instrument industry.


## 6 Acknowledgements

This paper is supported by the national key research and development Plan (2020YFB1713205) “Regional network collaborative manufacturing service technology and platform development and application demonstration”.


## 7 References


[1] Xiao, Y., Zheng, S., Shi, J. et al.: 'Knowledge graph-based manufacturing process planning: A state-of-the-art review', Journal of Intelligent Manufacturing， 2023, 70, pp. 417–435

[2] Dong, J., Wang, J., Chen, S.: 'Knowledge graph construction based on knowledge enhanced word embedding

model in manufacturing domain', Journal of Intelligent and Fuzzy Systems, 2021, 41, (2), pp. 3603–3613

[3] Lu, Y., Asghar, M.: 'Semantic communications between distributed cyber-physical systems towards collaborative automation for smart manufacturing', Journal of Manufacturing Systems, 2020, 55, pp. 348–359

[4] Wen, P., Ma, Y., Wang, R.: 'Systematic knowledge modeling and extraction methods for manufacturing process planning based on knowledge graph', Advanced Engineering Informatics, 2023, 58, pp. 102172

[5] Jiang, Z., Xie, B., Zhu, S., et al.: 'Dynamic data flow-driven knowledge graph construction method for remanufacturing disassembly process', Computer Integrated Manufacturing Systems, 2024, 30, (3), pp. 879–892

[6] Guan, K., Du, L., Yang, X.: ' Extraction and Processing for Knowledge Graph of Welding Manufacturing', IEEE Access, 2022, 10, pp. 103089–103098

[7] Zhang, X.: 'Research On Enterprise Knowledge Unified Retrieval Based On Industrial Big Data', 2018 Sixth International Conference on Advanced Cloud and Big Data, LanZhou, China, Aug 2018, pp. 130–134

[8] Cai, C.， Jiang, J., Wu, H., et al.: 'Research on knowledge graph-driven equipment fault diagnosis method for intelligent manufacturing', International Journal of Advanced Manufacturing Technology, 2024, 130, (9-10), pp. 4649–4662

[9] Wu, Y., Liu, F., Wan, L., et al.: 'Intelligent Fault Diagnostic Model for Industrial Equipment Based on Multimodal Knowledge Graph', IEEE Sensor Journal, 2023, 23, (21), pp. 26269–26278

[10] Manning, C. D., Surdeanu, M., Bauer, J., et al.: 'The Stanford CoreNLP Natural Language Processing Toolkit', 52nd Annual Meeting of the Association-for-Computational-Linguistics (ACL), Baltimore, USA, Jun 2014, pp. 55–60

[11] Zhang, K., Chen, W., Zai, H., et al.: 'A Construction Method of Electric Power Professional Domain Corpus Based on Multi-model Collaboration', 4th Asia Energy and Electrical Engineering Symposium (AEEES), Chengdu, China, Mar 2022, pp. 301–308